

INFORMATION SHEET

90nm COMPLETE DESIGN KIT

The Cadence 90nm generic Complete Design Kit (CDK) includes a standard cell library, IO cell library and process design kit (PDK). The CDK does contain proprietary Cadence intellectual property, but does not include any third part proprietary information.

The 90nm CDK is being developed to support the Virtuoso Flow Solutions 2H04 Ether and Raptor projects. The 90nm CDK and these projects will be used to develop flows, demos and workshops as follows:

Virtuoso Demos/Workshops	CDK090 Support
Virtuoso AMS Block Authoring	Yes
Virtuoso AMS Top-Down Sim.	Yes
Virtuoso AMS Chip Finishing	Yes
Virtuoso RFIC Flow	No

This information sheet (Version 1.0) describes the contents and status at the beginning of Q404. A list of frequently asked questions is included at the end of this document.

Table 1 Standard Cell and IO Views

Description	Extension	Std Cell gsclib090	I/O Cells giolib090	Comments
Netlist Data				
- Verilog Macro	*.v	Yes	No	Each cell has a Verilog macro definition
- Spice Netlist	*.spi	Yes	No	Each cell has a Spice netlist
- Extracted Netlist	*.cir	No	No	Netlist can be extracted from layout
CDB Library Data				
- DFII Technology File	*.tf	N/A	N/A	Included in GPDK090 process design kit library
- DFII Schematic	schematic view	Yes	Yes	Schematics were auto-generated
- DFII Symbol	symbol view	Yes	No	Symbols were all auto-generated
- DFII Layout	layout view	Yes	Yes	Each cell has a manual layout
- DFII Abstract	abstract view	Yes	No	IO cell abstracts will be generated NLT Feb '05
OA 2.0 Library Data				
- DFII Technology File	*.tf	N/A	N/A	Included in GPDK090 process design kit library
- Schematic	schematic view	Yes	No	Converted from CDBA to OA; IO plan is Feb '05
- Symbol	symbol view	Yes	No	Converted from CDBA to OA; IO plan is Feb '05
- Layout	layout view	Yes	No	Converted from CDBA to OA; IO plan is Feb '05
- DFII Abstract	abstract view	Yes	No	Converted from CDBA to OA; IO plan is Feb '05
Archival & Exchange Formats				
- EDIF Symbols	*.edf	No	No	EDIF is not used in the Cadence flows to date
- GDS	*.gds	No	No	Can be generated from Virtuoso layout views
- LEF abstracts	*.lef	Yes	Yes	Technology LEF supplied with gpdk090
- Antenna LEF	*_ant.lef	Yes	No	Not needed or planned for Ether/Raptor projects
Timing Data				
- Liberty format	*.lib	No	No	Refer to Table 2
- TLF format	*.tlf	No	No	Refer to Table 2
Other Cadence Tool Data				
- Signal Storm Library	*.alf	No	No	Experimental, not used in current flows
- Celtic	*.cdB	No	No	Experimental, not used in current flows

Timing data availability for the Standard Cells and IOs is listed in Table 2. It's important to note that the Raptor and Ether projects do not have any timing requirements and as a result the data has not been tested or validated.

Table 2 Standard Cell and IO Characterization

Characterization Table	Liberty	TLF	Comments
Standard Cells			
- Slow/Slow 125C 0.9V	Yes	No	Not needed or planned for Ether/Raptor projects
- Fast/Fast -40C 1.1V	No	No	Not needed or planned for Ether/Raptor projects
- Typ/Typ 25C 1.0V	No	No	Not needed or planned for Ether/Raptor projects
I/O Cells			
- Slow/Slow 125C 0.9V & 2.25V	No	No	Not needed or planned for Ether/Raptor projects
- Fast/Fast -40C 1.1V & 2.75V	No	No	Not needed or planned for Ether/Raptor projects
- Typ/Typ 25C 1.0V & 2.5V	No	No	Not needed or planned for Ether/Raptor projects

A list of the standard cells and IOs are provided separately in Tables 4 and 6.

Table 3 Process Design Kit (PDK)

Description	Extension	Files in PDK	Comments
Technology files			
- DFII Technology file (binary, ASCII)	*.tf, *.cds	Yes	Techfile objects needed an RF flow are not planned
- DFII Display.drf file (ASCII)	*.drf	Yes	Techfile objects needed an RF flow are not planned
- LEF Technology file (ASCII)	*.lef	Yes	
DFII CDB & OA 2.0 Pcells			
- mosfet (pmos, nmos)(lp, hv, s1v)	(in tf)	Yes	
- bjt (vpnp)	(in tf)	Yes	
- diode	(in tf)	Yes	
- moscap	(in tf)	Yes	
- poly resistor	(in tf)	Yes	
- symbolic (vias, contacts)	(in tf)	Yes	
- mimcap	(in tf)	No	Not needed or planned for the Ether/Raptor projects
- rf mosfets (pmos, nmos)	(in tf)	No	Not needed or planned for the Ether/Raptor projects
- varactor	(in tf)	No	Not needed or planned for the Ether/Raptor projects
- nwell resistor	(in tf)	No	Not needed or planned for the Ether/Raptor projects
- inductor	(in tf)	No	Not needed or planned for the Ether/Raptor projects
Simulation Models			
- mosfet (pmos, nmos)(lp, hv, s1v)	*.scs	Yes	Refer to Table 3a, Simulation
- bjt (vpnp)	*.scs	Yes	Refer to Table 3a, Simulation
- diode	*.scs	Yes	Refer to Table 3a, Simulation
- moscap	*.scs	Yes	Refer to Table 3a, Simulation
- poly resistor	*.scs	Yes	Refer to Table 3a, Simulation
- mimcap	*.scs	No	Not needed or planned for the Ether/Raptor projects
- rf mosfets (pmos, nmos)	*.scs	No	Not needed or planned for the Ether/Raptor projects
- varactor	*.scs	No	Not needed or planned for the Ether/Raptor projects
- nwell resistor	*.scs	No	Not needed or planned for the Ether/Raptor projects
- inductor	*.scs	No	Not needed or planned for the Ether/Raptor projects

Table 3 (continued) Process Design Kit (PDK)

Description	Extension	Files in PDK	Comments
Physical Verification Rule Decks			
- Diva DRC	*.rul	Yes	Antenna, density, drc
- Diva LVS	*.rul	Yes	Extraction, lvs
- Assura DRC	*.rul	Yes	Antenna, density, drc
- Assura LVS	*.rul	Yes	Extraction, Compare, LVS
- Assura RCX	*.rcx	Yes	proc, cap2d, capgen command rule
- Assura HF	?	No	Not needed or planned for the Ether/Raptor projects
- Assura RF	*.rul	No	Not needed or planned for the Ether/Raptor projects
- Calibre DRC/LVS/RCX	?	No	Not needed or planned for the Ether/Raptor projects
- Hercules DRC/LVS/RCX	?	No	Not needed or planned for the Ether/Raptor projects
Archival & Exchange Files			
- GDSII	streamLayer.Map	Yes	One layer-mapping file for both streamIn & streamOut
Other Cadence Tool Data			
- NeoCircuit	NeoDevice	Yes	Not tested; will be tested in Ether project NLT Jan '05
- NeoCell	NeoConfig	Yes	Not tested; will be tested in Ether project NTL Jan '05
- Virtuoso Layout Optimize	*.qtt	Yes	NeoCell 3.4 and later
- Virtuoso Chip Assembly Router	icc.rules	Yes	Not tested; will be tested in Ether project NTL Jan '05
- VoltageStorm (VSPE)	*.cl	Yes	Not tested; will be tested in Ether project NTL Jan '05
- VoltageStorm (VAVO)	*emData	Yes	Not tested; will be tested in Ether project NTL Jan '05
- Fire & Ice QX	*.ict, *.layermap	Yes	Experimental, not used in current flows
- SNA	*.cds, *.enc	Yes	Experimental, not used in current flows
- SOCE	*.capTbl	Yes	Experimental, not used in current flows

Table 3a Simulation

Simulation Models	BSIM3	Corners	Statistical (Mismatch)	Statistical (Process Variation)
- mosfet (pmos, nmos)(lp, hv, s1v)	Yes	FF/SS/TT/FS/SF	No	No
- rf mosfets (pmos, nmos)	No	FF/SS/TT/FS/SF	No	No
- bjt (vpnp)	Yes	FF/SS/TT/FS/SF	No	No
- diode	Yes	FF/SS/TT/FS/SF	No	No
- mimcap	Yes	FF/SS/TT/FS/SF	No	No
- moscap	Yes	FF/SS/TT/FS/SF	No	No
- varactor	Yes	FF/SS/TT/FS/SF	No	No
- poly resistor	Yes	FF/SS/TT/FS/SF	No	No
- nwell resistor	Yes	FF/SS/TT/FS/SF	No	No
- inductor	Yes	FF/SS/TT/FS/SF	No	No

The generic 90nm IO power requirements are 1.0 volt for core power and 2.5 volts for ESD & IO drive. The drive strength is fixed at 2ma and is non-programmable. A list of the available IO cells is provided in Table 4.

Table 4 IO Cell Features

Cell Name	Description	non-Stagger bond	Stagger bond	HVT	non-HVT	ESD Spec (KV & V)
I/O FEED CELLS						
- PFEED0005	Feed cell, width 0.0005 um (1 manufacturing unit wide)	Yes	No	No	Yes	N/A
- PFEED001	Feed cell, width 0.001 um wide	Yes	No	No	Yes	N/A
- PFEED01	Feed cell, width 0.01 um wide	Yes	No	No	Yes	N/A
- PFEED1	Feed cell, width 1 um wide	Yes	No	No	Yes	N/A
- PFEED3	Feed cell, width 3 um wide	Yes	No	No	Yes	N/A
- PFEED3x	Isolate PVDD/PVSS domain from other PVDD/PVSS domains	Yes	No	No	Yes	N/A
- PFEED5	Feed cell, width 5 um wide	Yes	No	No	Yes	N/A
- PFEED10	Feed cell, width 10 um wide	Yes	No	No	Yes	N/A
- PFEED60	Feed cell, width 60 um wide	Yes	No	No	Yes	N/A
- PCORNER	Corner cell	Yes	No	No	Yes	N/A
I/O POWER CELLS						
- PVDD	core power (1.0 volt)	Yes	No	No	Yes	Untested
- PVDD25	core power (2.0 volt)	Yes	No	No	Yes	Untested
- PVSS	core ground	Yes	No	No	Yes	Untested
- PVDDPAD	2.5 volt IO & 2.5 volt ESD pad power	Yes	No	No	Yes	Untested
- PVSSPAD	IO pad & ESD ground	Yes	No	No	Yes	Untested
I/O DIGITAL CELLS						
- PDB	PAD, Bidirectional	Yes	No	No	Yes	Untested
- PDI	PAD, Input	Yes	No	No	Yes	Untested
- PDO	Pad, Output (output 2ma)	Yes	No	No	Yes	Untested
- PDOZ	Pad, Output, (output 2ma) Tri-state high impedance	Yes	No	No	Yes	Untested
- PDIS	Pad, Schmitt-Trigger Input	No	No	No	No	Untested
- PDBS	Pad, Bidirectional (input/output 2ma) Schmitt-Trigger Input	No	No	No	No	Untested
I/O ANALOG CELLS						
- PANALOG	Pad, Analog (range 0 to 2.5 volts)	Yes	No	No	Yes	Untested
I/O RF CELLS						
	Not required for Ether and Raptor projects	No	No	No	No	Untested

Table 5 Standard Cells

Cell Family	Cell Type	Drive Strength	Prefix List
And	52	1,2,4,6,8,L	AND(18), NAND(34)
Or	64	1,2,4,6,8,L	OR(18), NOR(34), XOR(6), XNOR(6)
Inverter	10	1,2,3,4,6,8,12,16,20,L	INV(10)
Buffer/Tri-State-Buffer w/ & w/o enable/inverter	18	1,2,3,4,6,8,12,16,20,L	BUF(8), TBUF(10)
Clock Buffer/Inverted Clock Buffer/Gated Clock Latch	33	1,2,3,4,6,8,12,16,20	CLKBUF(8),CLKAND(6), CLKINV(9), CLKXOR(4), CLKMX(6)
AOI/OAI/AO/OA	96	1,2,4,L	AOI(40), OAI(40), AO(8), OA(8)
Half/Full Adder	12	1,2,4,L	ADDF(8), ADDH(4)
Mux with/without inverted output	28	1,2,4,6,8,L	MX(28)
DFF: pos-edge, neg-edge, async/sync R/S, w/o R/S	44	1,2,4,8,L	DFF(44)
Enable DFF; async/sync R, w/o R	16	1,2,4,8,L	EDFF(12), MDFF(4)
Scan version of all DFFs	60	1,2,4,8,L	SDFF(44), SEDFF(12), SMDFF(4)
Latch: active high/low enable, async R/S	32	1,2,3,4,6,8,12,16,20,L	TLAT(32)
Balanced Rise/Fall INV/NAND/AND/MUX/XOR	2	2,4	BMXI(2)
Dlay/Tie-High/Tie-Low	10	1,4, HI, LO	DLY(8), TIE(2)
Filler Cells	7	1,2,4,8,16,32,64	FILL(7)
Antenna cell/Decoupling cells	1	1	ANT(1)
Other cells	4	1	HOLD(1), VDD(1), VSS(1), ACH(1)
Total Cell Number	489		30 basic types

Table 6 List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
AND2X1/	AND logic gate	OR2X1/	OR logic gate
AND2X2/	AND logic gate	OR2X2/	OR logic gate
AND2X4/	AND logic gate	OR2X4/	OR logic gate
AND2X6/	AND logic gate	OR2X6/	OR logic gate
AND2X8/	AND logic gate	OR2X8/	OR logic gate
AND2XL/	AND logic gate	OR2XL/	OR logic gate
AND3X1/	AND logic gate	OR3X1/	OR logic gate
AND3X2/	AND logic gate	OR3X2/	OR logic gate
AND3X4/	AND logic gate	OR3X4/	OR logic gate
AND3X6/	AND logic gate	OR3X6/	OR logic gate
AND3X8/	AND logic gate	OR3X8/	OR logic gate
AND3XL/	AND logic gate	OR3XL/	OR logic gate
AND4X1/	AND logic gate	OR4X1/	OR logic gate
AND4X2/	AND logic gate	OR4X2/	OR logic gate
AND4X4/	AND logic gate	OR4X4/	OR logic gate
AND4X6/	AND logic gate	OR4X6/	OR logic gate
AND4X8/	AND logic gate	OR4X8/	OR logic gate
AND4XL/	AND logic gate	OR4XL/	OR logic gate
18		18	
NAND2BX1/	NOT AND logic gate	NOR2BX1/	NOT OR logic gate
NAND2BX2/	NOT AND logic gate	NOR2BX2/	NOT OR logic gate
NAND2BX4/	NOT AND logic gate	NOR2BX4/	NOT OR logic gate
NAND2BXL/	NOT AND logic gate	NOR2BXL/	NOT OR logic gate
NAND2X1/	NOT AND logic gate	NOR2X1/	NOT OR logic gate
NAND2X2/	NOT AND logic gate	NOR2X2/	NOT OR logic gate
NAND2X4/	NOT AND logic gate	NOR2X4/	NOT OR logic gate
NAND2X6/	NOT AND logic gate	NOR2X6/	NOT OR logic gate
NAND2X8/	NOT AND logic gate	NOR2X8/	NOT OR logic gate
NAND2XL/	NOT AND logic gate	NOR2XL/	NOT OR logic gate
NAND3BX1/	NOT AND logic gate	NOR3BX1/	NOT OR logic gate
NAND3BX2/	NOT AND logic gate	NOR3BX2/	NOT OR logic gate
NAND3BX4/	NOT AND logic gate	NOR3BX4/	NOT OR logic gate
NAND3BXL/	NOT AND logic gate	NOR3BXL/	NOT OR logic gate
NAND3X1/	NOT AND logic gate	NOR3X1/	NOT OR logic gate
NAND3X2/	NOT AND logic gate	NOR3X2/	NOT OR logic gate
NAND3X4/	NOT AND logic gate	NOR3X4/	NOT OR logic gate
NAND3X6/	NOT AND logic gate	NOR3X6/	NOT OR logic gate
NAND3X8/	NOT AND logic gate	NOR3X8/	NOT OR logic gate
NAND3XL/	NOT AND logic gate	NOR3XL/	NOT OR logic gate
NAND4BBX1/	NOT AND logic gate	NOR4BBX1/	NOT OR logic gate
NAND4BBX2/	NOT AND logic gate	NOR4BBX2/	NOT OR logic gate
NAND4BBX4/	NOT AND logic gate	NOR4BBX4/	NOT OR logic gate
NAND4BBXL/	NOT AND logic gate	NOR4BBXL/	NOT OR logic gate
NAND4BX1/	NOT AND logic gate	NOR4BX1/	NOT OR logic gate
NAND4BX2/	NOT AND logic gate	NOR4BX2/	NOT OR logic gate
NAND4BX4/	NOT AND logic gate	NOR4BX4/	NOT OR logic gate
NAND4BXL/	NOT AND logic gate	NOR4BXL/	NOT OR logic gate
NAND4X1/	NOT AND logic gate	NOR4X1/	NOT OR logic gate
NAND4X2/	NOT AND logic gate	NOR4X2/	NOT OR logic gate
NAND4X4/	NOT AND logic gate	NOR4X4/	NOT OR logic gate
NAND4X6/	NOT AND logic gate	NOR4X6/	NOT OR logic gate
NAND4X8/	NOT AND logic gate	NOR4X8/	NOT OR logic gate
NAND4XL/	NOT AND logic gate	NOR4XL/	NOT OR logic gate
34		34	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
XOR2X1/	EXCLUSIVE OR logic gate	CLKAND2X12/	CLOCKED AND
XOR2X2/	EXCLUSIVE OR logic gate	CLKAND2X2/	CLOCKED AND
XOR2X4/	EXCLUSIVE OR logic gate	CLKAND2X3/	CLOCKED AND
XOR2XL/	EXCLUSIVE OR logic gate	CLKAND2X4/	CLOCKED AND
XOR3X1/	EXCLUSIVE OR logic gate	CLKAND2X6/	CLOCKED AND
XOR3XL/	EXCLUSIVE OR logic gate	CLKAND2X8/	CLOCKED AND
6		6	
XNOR2X1/	EXCLUSIVE NOT OR logic gate	CLKINVX1/	CLOCKED INVERTER
XNOR2X2/	EXCLUSIVE NOT OR logic gate	CLKINVX12/	CLOCKED INVERTER
XNOR2X4/	EXCLUSIVE NOT OR logic gate	CLKINVX16/	CLOCKED INVERTER
XNOR2XL/	EXCLUSIVE NOT OR logic gate	CLKINVX2/	CLOCKED INVERTER
XNOR3X1/	EXCLUSIVE NOT OR logic gate	CLKINVX20/	CLOCKED INVERTER
XNOR3XL/	EXCLUSIVE NOT OR logic gate	CLKINVX3/	CLOCKED INVERTER
6		CLKINVX4/	CLOCKED INVERTER
INVX1	INVERTER	CLKINVX6/	CLOCKED INVERTER
INVX12	INVERTER	CLKINVX8/	CLOCKED INVERTER
INVX16	INVERTER	9	
INVX2	INVERTER	CLKXOR2X1/	CLOCKED EXCLUSIVE OR
INVX20	INVERTER	CLKXOR2X2/	CLOCKED EXCLUSIVE OR
INVX3	INVERTER	CLKXOR2X4/	CLOCKED EXCLUSIVE OR
INVX4	INVERTER	CLKXOR2X8/	CLOCKED EXCLUSIVE OR
INVX6	INVERTER	4	
INVX8	INVERTER	CLKMX2X12/	CLOCKED MULTIPLEXOR
INVXL	INVERTER	CLKMX2X2/	CLOCKED MULTIPLEXOR
10		CLKMX2X3/	CLOCKED MULTIPLEXOR
BUFX12/	BUFFER	CLKMX2X4/	CLOCKED MULTIPLEXOR
BUFX16/	BUFFER	CLKMX2X6/	CLOCKED MULTIPLEXOR
BUFX2/	BUFFER	CLKMX2X8/	CLOCKED MULTIPLEXOR
BUFX20/	BUFFER	6	
BUFX3/	BUFFER	OA21X1/	OR-AND logic
BUFX4/	BUFFER	OA21X2/	OR-AND logic
BUFX6/	BUFFER	OA21X4/	OR-AND logic
BUFX8/	BUFFER	OA21XL/	OR-AND logic
8		OA22X1/	OR-AND logic
TBUFX1/	TRI-STATE BUFFER	OA22X2/	OR-AND logic
TBUFX12/	TRI-STATE BUFFER	OA22X4/	OR-AND logic
TBUFX16/	TRI-STATE BUFFER	OA22XL/	OR-AND logic
TBUFX2/	TRI-STATE BUFFER	8	
TBUFX20/	TRI-STATE BUFFER	AO21X1/	AND-OR logic
TBUFX3/	TRI-STATE BUFFER	AO21X2/	AND-OR logic
TBUFX4/	TRI-STATE BUFFER	AO21X4/	AND-OR logic
TBUFX6/	TRI-STATE BUFFER	AO21XL/	AND-OR logic
TBUFX8/	TRI-STATE BUFFER	AO22X1/	AND-OR logic
TBUFXL/	TRI-STATE BUFFER	AO22X2/	AND-OR logic
10		AO22X4/	AND-OR logic
CLKBUFX12/	CLOCKED BUFFER	AO22XL/	AND-OR logic
CLKBUFX16/	CLOCKED BUFFER	8	
CLKBUFX2/	CLOCKED BUFFER		
CLKBUFX20/	CLOCKED BUFFER		
CLKBUFX3/	CLOCKED BUFFER		
CLKBUFX4/	CLOCKED BUFFER		
CLKBUFX6/	CLOCKED BUFFER		
CLKBUFX8/	CLOCKED BUFFER		
8			

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
AOI211X1/	AND-OR-INVERT logic	OAI211X1/	OR-AND-INVERT logic
AOI211X2/	AND-OR-INVERT logic	OAI211X2/	OR-AND-INVERT logic
AOI211X4/	AND-OR-INVERT logic	OAI211X4/	OR-AND-INVERT logic
AOI211XL/	AND-OR-INVERT logic	OAI211XL/	OR-AND-INVERT logic
AOI21X1/	AND-OR-INVERT logic	OAI21X1/	OR-AND-INVERT logic
AOI21X2/	AND-OR-INVERT logic	OAI21X2/	OR-AND-INVERT logic
AOI21X4/	AND-OR-INVERT logic	OAI21X4/	OR-AND-INVERT logic
AOI21XL/	AND-OR-INVERT logic	OAI21XL/	OR-AND-INVERT logic
AOI221X1/	AND-OR-INVERT logic	OAI221X1/	OR-AND-INVERT logic
AOI221X2/	AND-OR-INVERT logic	OAI221X2/	OR-AND-INVERT logic
AOI221X4/	AND-OR-INVERT logic	OAI221X4/	OR-AND-INVERT logic
AOI221XL/	AND-OR-INVERT logic	OAI221XL/	OR-AND-INVERT logic
AOI222X1/	AND-OR-INVERT logic	OAI222X1/	OR-AND-INVERT logic
AOI222X2/	AND-OR-INVERT logic	OAI222X2/	OR-AND-INVERT logic
AOI222X4/	AND-OR-INVERT logic	OAI222X4/	OR-AND-INVERT logic
AOI222XL/	AND-OR-INVERT logic	OAI222XL/	OR-AND-INVERT logic
AOI22X1/	AND-OR-INVERT logic	OAI22X1/	OR-AND-INVERT logic
AOI22X2/	AND-OR-INVERT logic	OAI22X2/	OR-AND-INVERT logic
AOI22X4/	AND-OR-INVERT logic	OAI22X4/	OR-AND-INVERT logic
AOI22XL/	AND-OR-INVERT logic	OAI22XL/	OR-AND-INVERT logic
AOI2BB1X1/	AND-OR-INVERT logic	OAI2BB1X1/	OR-AND-INVERT logic
AOI2BB1X2/	AND-OR-INVERT logic	OAI2BB1X2/	OR-AND-INVERT logic
AOI2BB1X4/	AND-OR-INVERT logic	OAI2BB1X4/	OR-AND-INVERT logic
AOI2BB1XL/	AND-OR-INVERT logic	OAI2BB1XL/	OR-AND-INVERT logic
AOI2BB2X1/	AND-OR-INVERT logic	OAI2BB2X1/	OR-AND-INVERT logic
AOI2BB2X2/	AND-OR-INVERT logic	OAI2BB2X2/	OR-AND-INVERT logic
AOI2BB2X4/	AND-OR-INVERT logic	OAI2BB2X4/	OR-AND-INVERT logic
AOI2BB2XL/	AND-OR-INVERT logic	OAI2BB2XL/	OR-AND-INVERT logic
AOI31X1/	AND-OR-INVERT logic	OAI31X1/	OR-AND-INVERT logic
AOI31X2/	AND-OR-INVERT logic	OAI31X2/	OR-AND-INVERT logic
AOI31X4/	AND-OR-INVERT logic	OAI31X4/	OR-AND-INVERT logic
AOI31XL/	AND-OR-INVERT logic	OAI31XL/	OR-AND-INVERT logic
AOI32X1/	AND-OR-INVERT logic	OAI32X1/	OR-AND-INVERT logic
AOI32X2/	AND-OR-INVERT logic	OAI32X2/	OR-AND-INVERT logic
AOI32X4/	AND-OR-INVERT logic	OAI32X4/	OR-AND-INVERT logic
AOI32XL/	AND-OR-INVERT logic	OAI32XL/	OR-AND-INVERT logic
AOI33X1/	AND-OR-INVERT logic	OAI33X1/	OR-AND-INVERT logic
AOI33X2/	AND-OR-INVERT logic	OAI33X2/	OR-AND-INVERT logic
AOI33X4/	AND-OR-INVERT logic	OAI33X4/	OR-AND-INVERT logic
AOI33XL/	AND-OR-INVERT logic	OAI33XL/	OR-AND-INVERT logic
40		40	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
ADDFHX1/	FULL ADDER	DFFHQX1/	D-FLIP-FLOP
ADDFHX2/	FULL ADDER	DFFHQX2/	D-FLIP-FLOP
ADDFHX4/	FULL ADDER	DFFHQX4/	D-FLIP-FLOP
ADDFHXL/	FULL ADDER	DFFHQX8/	D-FLIP-FLOP
ADDFX1/	FULL ADDER	DFFNSRX1/	D-FLIP-FLOP
ADDFX2/	FULL ADDER	DFFNSRX2/	D-FLIP-FLOP
ADDFX4/	FULL ADDER	DFFNSRX4/	D-FLIP-FLOP
ADDFXL/	FULL ADDER	DFFNSRXL/	D-FLIP-FLOP
8		DFFQX1/	D-FLIP-FLOP
ADDHX1/	HALF ADDER	DFFQX2/	D-FLIP-FLOP
ADDHX2/	HALF ADDER	DFFQX4/	D-FLIP-FLOP
ADDHX4/	HALF ADDER	DFFQXL/	D-FLIP-FLOP
ADDHXL/	HALF ADDER	DFFRHQX1/	D-FLIP-FLOP
4		DFFRHQX2/	D-FLIP-FLOP
MX2X1/	MULTIPLEXOR	DFFRHQX4/	D-FLIP-FLOP
MX2X2/	MULTIPLEXOR	DFFRHQX8/	D-FLIP-FLOP
MX2X4/	MULTIPLEXOR	DFFRX1/	D-FLIP-FLOP
MX2X6/	MULTIPLEXOR	DFFRX2/	D-FLIP-FLOP
MX2X8/	MULTIPLEXOR	DFFRX4/	D-FLIP-FLOP
MX2XL/	MULTIPLEXOR	DFFRXL/	D-FLIP-FLOP
MX3X1/	MULTIPLEXOR	DFFSHQX1/	D-FLIP-FLOP
MX3X2/	MULTIPLEXOR	DFFSHQX2/	D-FLIP-FLOP
MX3X4/	MULTIPLEXOR	DFFSHQX4/	D-FLIP-FLOP
MX3XL/	MULTIPLEXOR	DFFSHQX8/	D-FLIP-FLOP
MX4X1/	MULTIPLEXOR	DFFSRHQX1/	D-FLIP-FLOP
MX4X2/	MULTIPLEXOR	DFFSRHQX2/	D-FLIP-FLOP
MX4X4/	MULTIPLEXOR	DFFSRHQX4/	D-FLIP-FLOP
MX4XL/	MULTIPLEXOR	DFFSRHQX8/	D-FLIP-FLOP
MXI2X1/	MULTIPLEXOR	DFFSRX1/	D-FLIP-FLOP
MXI2X2/	MULTIPLEXOR	DFFSRX2/	D-FLIP-FLOP
MXI2X4/	MULTIPLEXOR	DFFSRX4/	D-FLIP-FLOP
MXI2X6/	MULTIPLEXOR	DFFSRXL/	D-FLIP-FLOP
MXI2X8/	MULTIPLEXOR	DFFSX1/	D-FLIP-FLOP
MXI2XL/	MULTIPLEXOR	DFFSX2/	D-FLIP-FLOP
MXI3X1/	MULTIPLEXOR	DFFSX4/	D-FLIP-FLOP
MXI3X2/	MULTIPLEXOR	DFFSXL/	D-FLIP-FLOP
MXI3X4/	MULTIPLEXOR	DFFTRX1/	D-FLIP-FLOP
MXI3XL/	MULTIPLEXOR	DFFTRX2/	D-FLIP-FLOP
MXI4X1/	MULTIPLEXOR	DFFTRX4/	D-FLIP-FLOP
MXI4X2/	MULTIPLEXOR	DFFTRXL/	D-FLIP-FLOP
MXI4X4/	MULTIPLEXOR	DFFX1/	D-FLIP-FLOP
MXI4XL/	MULTIPLEXOR	DFFX2/	D-FLIP-FLOP
28		DFFX4/	D-FLIP-FLOP
BMXIX2/	MULTIPLEXOR	DFFXL/	D-FLIP-FLOP
BMXIX4/	MULTIPLEXOR	44	
2			

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
EDFFHQX1/	ENABLE D-FLIP-FLOP	SDFFHQX1/	SCAN D-FLIP-FLOP
EDFFHQX2/	ENABLE D-FLIP-FLOP	SDFFHQX2/	SCAN D-FLIP-FLOP
EDFFHQX4/	ENABLE D-FLIP-FLOP	SDFFHQX4/	SCAN D-FLIP-FLOP
EDFFHQX8/	ENABLE D-FLIP-FLOP	SDFFHQX8/	SCAN D-FLIP-FLOP
EDFFTRX1/	ENABLE D-FLIP-FLOP	SDFFNSRX1/	SCAN D-FLIP-FLOP
EDFFTRX2/	ENABLE D-FLIP-FLOP	SDFFNSRX2/	SCAN D-FLIP-FLOP
EDFFTRX4/	ENABLE D-FLIP-FLOP	SDFFNSRX4/	SCAN D-FLIP-FLOP
EDFFTRXL/	ENABLE D-FLIP-FLOP	SDFFNSRXL/	SCAN D-FLIP-FLOP
EDFFX1/	ENABLE D-FLIP-FLOP	SDFFQX1/	SCAN D-FLIP-FLOP
EDFFX2/	ENABLE D-FLIP-FLOP	SDFFQX2/	SCAN D-FLIP-FLOP
EDFFX4/	ENABLE D-FLIP-FLOP	SDFFQX4/	SCAN D-FLIP-FLOP
EDFFXL/	ENABLE D-FLIP-FLOP	SDFFQXL/	SCAN D-FLIP-FLOP
12		SDFFRHQX1/	SCAN D-FLIP-FLOP
MDFFHQX1/	MUX D-FLIP-FLOP	SDFFRHQX2/	SCAN D-FLIP-FLOP
MDFFHQX2/	MUX D-FLIP-FLOP	SDFFRHQX4/	SCAN D-FLIP-FLOP
MDFFHQX4/	MUX D-FLIP-FLOP	SDFFRHQX8/	SCAN D-FLIP-FLOP
MDFFHQX8/		SDFFRX1/	SCAN D-FLIP-FLOP
4		SDFFRX2/	SCAN D-FLIP-FLOP
SEDFFHQX1/	SCAN ENABLE D-FLIP-FLOP	SDFFRX4/	SCAN D-FLIP-FLOP
SEDFFHQX2/	SCAN ENABLE D-FLIP-FLOP	SDFFRXL/	SCAN D-FLIP-FLOP
SEDFFHQX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX1/	SCAN D-FLIP-FLOP
SEDFFHQX8/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX2/	SCAN D-FLIP-FLOP
SEDFFTRX1/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX4/	SCAN D-FLIP-FLOP
SEDFFTRX2/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX8/	SCAN D-FLIP-FLOP
SEDFFTRX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX1/	SCAN D-FLIP-FLOP
SEDFFTRXL/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX2/	SCAN D-FLIP-FLOP
SEDFFX1/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX4/	SCAN D-FLIP-FLOP
SEDFFX2/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX8/	SCAN D-FLIP-FLOP
SEDFFX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSRX1/	SCAN D-FLIP-FLOP
SEDFFXL/	SCAN ENABLE D-FLIP-FLOP	SDFFSRX2/	SCAN D-FLIP-FLOP
12		SDFFSRX4/	SCAN D-FLIP-FLOP
SMDFFHQX1/	SCAN MUX D-FLIP-FLOP	SDFFSRXL/	SCAN D-FLIP-FLOP
SMDFFHQX2/	SCAN MUX D-FLIP-FLOP	SDFFSX1/	SCAN D-FLIP-FLOP
SMDFFHQX4/	SCAN MUX D-FLIP-FLOP	SDFFSX2/	SCAN D-FLIP-FLOP
SMDFFHQX8/	SCAN MUX D-FLIP-FLOP	SDFFSX4/	SCAN D-FLIP-FLOP
4		SDFFSXL/	SCAN D-FLIP-FLOP
		SDFFTRX1/	SCAN D-FLIP-FLOP
		SDFFTRX2/	SCAN D-FLIP-FLOP
		SDFFTRX4/	SCAN D-FLIP-FLOP
		SDFFTRXL/	SCAN D-FLIP-FLOP
		SDFFX1/	SCAN D-FLIP-FLOP
		SDFFX2/	SCAN D-FLIP-FLOP
		SDFFX4/	SCAN D-FLIP-FLOP
		SDFFXL/	SCAN D-FLIP-FLOP
		44	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
EDFFHQX1/	ENABLE D-FLIP-FLOP	SDFFHQX1/	SCAN D-FLIP-FLOP
EDFFHQX2/	ENABLE D-FLIP-FLOP	SDFFHQX2/	SCAN D-FLIP-FLOP
EDFFHQX4/	ENABLE D-FLIP-FLOP	SDFFHQX4/	SCAN D-FLIP-FLOP
EDFFHQX8/	ENABLE D-FLIP-FLOP	SDFFHQX8/	SCAN D-FLIP-FLOP
EDFFTRX1/	ENABLE D-FLIP-FLOP	SDFFNSRX1/	SCAN D-FLIP-FLOP
EDFFTRX2/	ENABLE D-FLIP-FLOP	SDFFNSRX2/	SCAN D-FLIP-FLOP
EDFFTRX4/	ENABLE D-FLIP-FLOP	SDFFNSRX4/	SCAN D-FLIP-FLOP
EDFFTRXL/	ENABLE D-FLIP-FLOP	SDFFNSRXL/	SCAN D-FLIP-FLOP
EDFFX1/	ENABLE D-FLIP-FLOP	SDFFQX1/	SCAN D-FLIP-FLOP
EDFFX2/	ENABLE D-FLIP-FLOP	SDFFQX2/	SCAN D-FLIP-FLOP
EDFFX4/	ENABLE D-FLIP-FLOP	SDFFQX4/	SCAN D-FLIP-FLOP
EDFFXL/	ENABLE D-FLIP-FLOP	SDFFQXL/	SCAN D-FLIP-FLOP
12		SDFFRHQX1/	SCAN D-FLIP-FLOP
MDFFHQX1/	MUX D-FLIP-FLOP	SDFFRHQX2/	SCAN D-FLIP-FLOP
MDFFHQX2/	MUX D-FLIP-FLOP	SDFFRHQX4/	SCAN D-FLIP-FLOP
MDFFHQX4/	MUX D-FLIP-FLOP	SDFFRHQX8/	SCAN D-FLIP-FLOP
MDFFHQX8/		SDFFRX1/	SCAN D-FLIP-FLOP
4		SDFFRX2/	SCAN D-FLIP-FLOP
SEDFFHQX1/	SCAN ENABLE D-FLIP-FLOP	SDFFRX4/	SCAN D-FLIP-FLOP
SEDFFHQX2/	SCAN ENABLE D-FLIP-FLOP	SDFFRXL/	SCAN D-FLIP-FLOP
SEDFFHQX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX1/	SCAN D-FLIP-FLOP
SEDFFHQX8/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX2/	SCAN D-FLIP-FLOP
SEDFFTRX1/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX4/	SCAN D-FLIP-FLOP
SEDFFTRX2/	SCAN ENABLE D-FLIP-FLOP	SDFFSHQX8/	SCAN D-FLIP-FLOP
SEDFFTRX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX1/	SCAN D-FLIP-FLOP
SEDFFTRXL/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX2/	SCAN D-FLIP-FLOP
SEDFFX1/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX4/	SCAN D-FLIP-FLOP
SEDFFX2/	SCAN ENABLE D-FLIP-FLOP	SDFFSRHQX8/	SCAN D-FLIP-FLOP
SEDFFX4/	SCAN ENABLE D-FLIP-FLOP	SDFFSRX1/	SCAN D-FLIP-FLOP
SEDFFXL/	SCAN ENABLE D-FLIP-FLOP	SDFFSRX2/	SCAN D-FLIP-FLOP
12		SDFFSRX4/	SCAN D-FLIP-FLOP
SMDFFHQX1/	SCAN MUX D-FLIP-FLOP	SDFFSRXL/	SCAN D-FLIP-FLOP
SMDFFHQX2/	SCAN MUX D-FLIP-FLOP	SDFFSX1/	SCAN D-FLIP-FLOP
SMDFFHQX4/	SCAN MUX D-FLIP-FLOP	SDFFSX2/	SCAN D-FLIP-FLOP
SMDFFHQX8/	SCAN MUX D-FLIP-FLOP	SDFFSX4/	SCAN D-FLIP-FLOP
4		SDFFSXL/	SCAN D-FLIP-FLOP
		SDFFTRX1/	SCAN D-FLIP-FLOP
		SDFFTRX2/	SCAN D-FLIP-FLOP
		SDFFTRX4/	SCAN D-FLIP-FLOP
		SDFFTRXL/	SCAN D-FLIP-FLOP
		SDFFX1/	SCAN D-FLIP-FLOP
		SDFFX2/	SCAN D-FLIP-FLOP
		SDFFX4/	SCAN D-FLIP-FLOP
		SDFFXL/	SCAN D-FLIP-FLOP
		44	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description	List of Standard Cells	Description
TLATNCAX12/	TRI-STATE LATCH	ANTENNA/	ANTENNA CELL
TLATNCAX16/	TRI-STATE LATCH	1	
TLATNCAX2/	TRI-STATE LATCH	HOLDX1/	MEMORY INPUT/OUTPUT LATCH
TLATNCAX20/	TRI-STATE LATCH	1	
TLATNCAX3/	TRI-STATE LATCH	VDD/	INHERITED CONNECTION POWER
TLATNCAX4/	TRI-STATE LATCH	1	
TLATNCAX6/	TRI-STATE LATCH	VSS/	INHERITED CONNECTION GROUND
TLATNCAX8/	TRI-STATE LATCH	1	
TLATNSRX1/	TRI-STATE LATCH	ACHCONX2	ACCUMULATOR
TLATNSRX2/	TRI-STATE LATCH	1	
TLATNSRX4/	TRI-STATE LATCH		
TLATNSRXL/	TRI-STATE LATCH		
TLATNTSCAX12/	TRI-STATE LATCH		
TLATNTSCAX16/	TRI-STATE LATCH		
TLATNTSCAX2/	TRI-STATE LATCH		
TLATNTSCAX20/	TRI-STATE LATCH		
TLATNTSCAX3/	TRI-STATE LATCH		
TLATNTSCAX4/	TRI-STATE LATCH		
TLATNTSCAX6/	TRI-STATE LATCH		
TLATNTSCAX8/	TRI-STATE LATCH		
TLATNX1/	TRI-STATE LATCH		
TLATNX2/	TRI-STATE LATCH		
TLATNX4/	TRI-STATE LATCH		
TLATNXL/	TRI-STATE LATCH		
TLATSRX1/	TRI-STATE LATCH		
TLATSRX2/	TRI-STATE LATCH		
TLATSRX4/	TRI-STATE LATCH		
TLATSRXL/	TRI-STATE LATCH		
TLATX1/	TRI-STATE LATCH		
TLATX2/	TRI-STATE LATCH		
TLATX4/	TRI-STATE LATCH		
TLATXL/	TRI-STATE LATCH		
32			
DLY1X1/	DELAY		
DLY1X4/	DELAY		
DLY2X1/	DELAY		
DLY2X4/	DELAY		
DLY3X1/	DELAY		
DLY3X4/	DELAY		
DLY4X1/	DELAY		
DLY4X4/	DELAY		
8			
TIEHI/	TIE HIGH		
TIELO/	TIE LOW		
2			
FILL1/	FILLER CELL		
FILL16/	FILLER CELL		
FILL2/	FILLER CELL		
FILL32/	FILLER CELL		
FILL4/	FILLER CELL		
FILL64/	FILLER CELL		
FILL8/	FILLER CELL		
7			

90nm CDK Access & Distribution

The 90nm CDK is currently in development. As a result, the design kit has not been tested completely with the Virtuoso Flow Solutions AMS and Chip Integration flows. These flows are expected to be complete in February 2005.

Any user needing access to the 90nm CDK prior to February 2005 for internal development purposes can contact John Gianni or Jeff Guy.

Due to the current development status, the 90nm CDK is not intended for distribution/release to Cadence customers or other 3rd parties at this time. Any requests for 3rd party distribution should be provided to Jeff Guy.

Frequently Asked Questions

- QUESTION **What is the purpose of the Cadence CDK090?**
 ANSWER *The Cadence 90nm complete design kit enables the 2H04 Ether and Raptor projects from which the AMS and Chip Integration custom IC flows, demos, & workshops will be created, tested, & delivered by internal Cadence groups.*
- QUESTION **Who is the main contact person for the Cadence CDK090?**
 ANSWER *For technical questions, contact John Gianni. All other questions for the Cadence CDK090 can be directed to Jeff Guy.*
- QUESTION **Does the Cadence CDK090 contain industry proprietary or non exportable data?**
 ANSWER *No; however the CDK090 is the property of Cadence Design Systems, and can only be used with permission.*
- QUESTION **What Cadence IP will be built using the CDK090 by Q404?**
 ANSWER *The Ethernet Phy and Raptor wired communications controller (includes Condor MPU, Parrot 8 port Ethernet Phy, & Hummingbird PLL)*
- QUESTION **Is a memory compiler part of the 90nm CDK?**
 ANSWER *No, memory block and associated design tasks are considered to be part of the design IP associated with a particular reference design. We do not have a VFS solution for designing memory blocks.*
- QUESTION **Are there plans for multi-Vt cells for low leakage and low-power design?**
 ANSWER *There are no plans at this time as these are not needed for the specific objectives.*
- QUESTION **Are there plans to support FPGA design?**
 ANSWER *There are no plans at this time as these are not needed for the specific objectives.*
- QUESTION **Is there an AMS Siminfo section in the CDF (PDK) for primitives??**
 ANSWER *Yes there is an AMS siminfo for each primitive in the cdf. Also, if using ADE SPICE Netlist Procedures (such as in SpectreDirect) these can be converted to AMS Netlist Procedures using the translator shipped with DFII.*
- QUESTION **Are the simulation models BSIM3 or BSIM4?**
 ANSWER *BSIM3; however, our BSIM3 models leakage currents are similar to those of BSIM4*

QUESTION **How would you compare the Cadence 90nm complete design kit with other industry kits?**

ANSWER *Though not completely tested, the process design kit is on par with industry standards; the standard cell library is only an implementation of the basic cells which meet the Ether/Raptor minimum needs; and the I/O library contains only the minimum necessary for Ether/Raptor needs.*

QUESTION **Does the Cadence CDK090 support the proper use of inherited connections?**

ANSWER *Yes; the PDK, standard cells, and I/O cells in the CDK fully support the Cadence-recommended inherited connections implementation.*